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Saveetha Institute of Medical And Technical Sciences
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Sharpening a Zoomed-In CCTV Face Frame

DSA0203-COMPUTER VISION WITH OPEN CV

Guided By:

Dr. NARENTIKUMAR
Reddy(192424230)

Presented By:

Y. Varun Kumar



Introduction



Description

Image sharpening improves the clarity of a zoomed CCTV face by enhancing edges and fine details.

Key Points

- Enhances blurry CCTV face images.
- Makes facial features more visible.
- Uses image filtering techniques.
- Improves edge sharpness.
- Useful before face recognition.



Objectives



- Read the CCTV face image.
- Zoom into the face region.
- Apply a sharpening filter.
- Display original and sharpened images.
- Improve image clarity.



Technologies Used



- **Image Cropping** – Extracts only the face region from the CCTV image.
- **Image Resizing (Zooming)** – Enlarges the cropped face for better visibility.
- **Convolution Filtering** – Applies a mathematical filter to the image.
- **Image Sharpening** – Enhances edges and fine facial details.
- **Kernel-Based Filtering** – Uses a sharpening kernel (mask) with filter2D().
- **Edge Enhancement** – Increases the contrast around edges to make features clearer.



Proposed Methodology

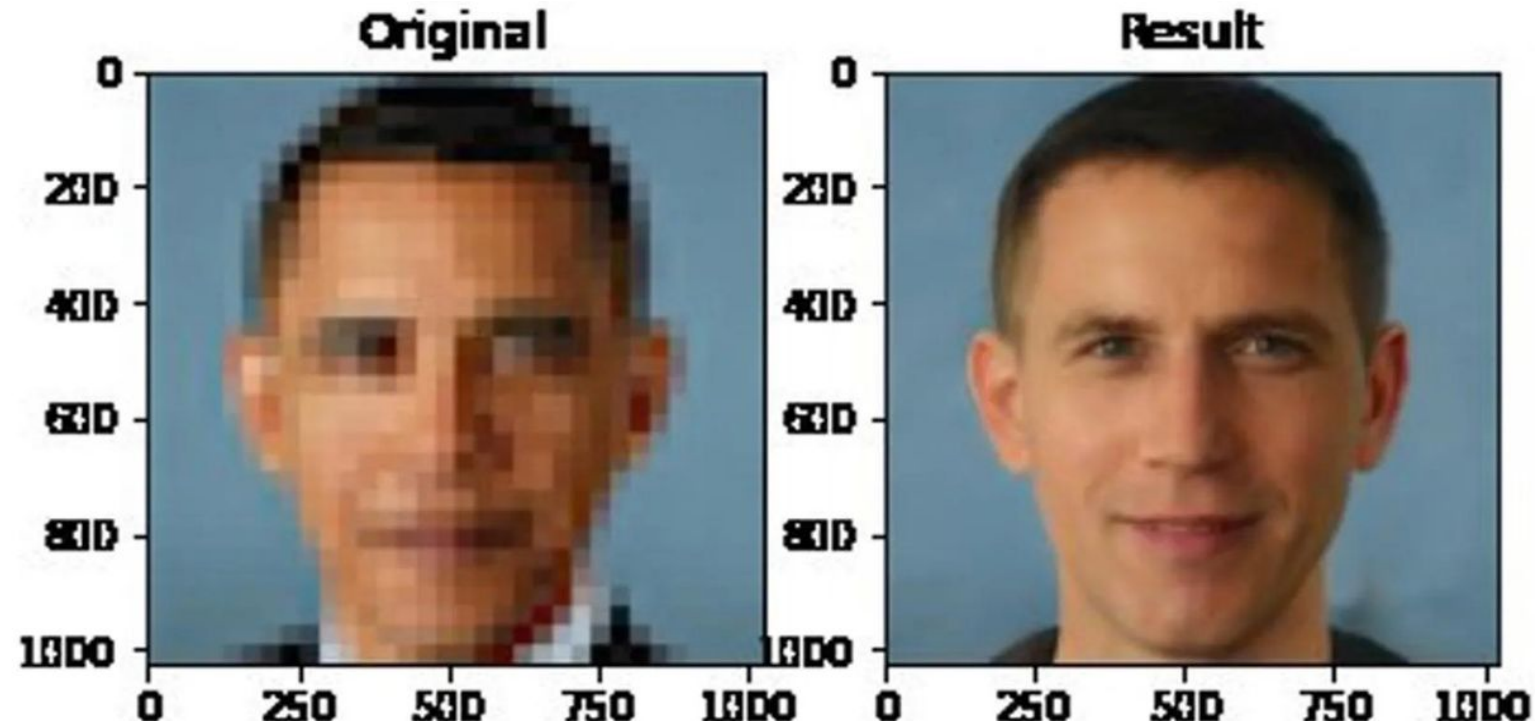


- Capture the CCTV image.
- Crop the face from the image.
- Zoom the face region.
- Apply a sharpening filter to enhance the image.
- Display and compare the original and sharpened images.

Output

Expected Result

- Original CCTV face frame.
- Zoomed face frame.
- Sharpened zoomed face with improved clarity.





Applications



- CCTV surveillance.
- Criminal investigation.
- Face recognition systems.
- Security monitoring.
- Forensic image enhancement.



Advantages



- Improves image quality.
- Enhances facial details.
- Easy to implement.
- Fast processing.
- Supports better recognition accuracy.

Conclusion

- Image sharpening significantly improves the clarity of zoomed-in CCTV face images.
- The sharpening filter enhances edges and highlights important facial details.
- It helps improve visibility for surveillance and security applications.
- The technique is simple, fast, and easy to implement using OpenCV.
- Sharpened images can improve the accuracy of face detection and recognition systems.
- Overall, image sharpening is an effective preprocessing step for enhancing CCTV footage before further analysis.

THANK
YOU